

Development of a Real-Time IoT Sensor System for Water Quality Monitoring Using Satellite Connectivity

Nnamdi Ugwuoke ¹ , Elizabeth Hart ², Mari Davies ² and Edore Akpokodje ^{1,*}

¹ Department of Computer Science, Aberystwyth University, Aberystwyth SY23 3DB, UK. nnu1@aber.ac.uk

² Department of Life Sciences, Aberystwyth University, Aberystwyth SY23 3DA, UK. mwd9@aber.ac.uk(M.D.); elh18@aber.ac.uk(E.H.).

* Correspondence: eta@aber.ac.uk;

Abstract

The growing demand for sustainable water resource management has intensified the need for real-time, continuous, and reliable monitoring solutions. Traditional methods, based on periodic manual sampling and ground-based data transmission, fail to capture short-term variations and are often infeasible in remote areas without communication infrastructure. This study presents the design, development, and field validation of an autonomous Internet of Things (IoT) system for real-time water quality monitoring, equipped with multi-parameter sensing capabilities and satellite-based data communication. The system measures key physicochemical indicators pH, nitrates, temperature, dissolved oxygen, electrical conductivity and ammonium, transmits data to a cloud-based platform using a low-power satellite network. Embedded edge processing enables anomaly detection and automatic alert generation, ensuring timely responses to contamination events. The proposed system demonstrates reliable data accuracy, energy efficiency, and communication stability during field deployment. Results indicate that the satellite-IoT approach effectively overcomes limitations of cellular and terrestrial networks, offering a scalable solution for continuous monitoring of isolated freshwater bodies. This work contributes to the advancement of IoT-enabled environmental monitoring through the integration of satellite communication, low-power design, and intelligent data analytics.

Keywords: IoT; water quality monitoring; satellite communication; environmental sensing; real-time data; smart sensor networks; edge computing; low-power design

0. Introduction

Access to clean water is fundamental to human health, biodiversity, and economic development. However, increasing industrial activity, agricultural runoff, and climate variability have intensified water pollution and resource stress.[5,6] Continuous monitoring of surface and groundwater quality is essential to detect contamination events and ensure compliance with environmental regulations.[10,11] Traditional water quality assessment methods rely on manual sampling and laboratory analysis, which are costly, labor-intensive, and temporally limited. Consequently, brief but ecologically significant pollution spikes often go undetected.

The rapid advancement of Internet of Things (IoT) technology has transformed environmental monitoring by enabling distributed, low-cost, and connected sensor systems. However, most existing IoT-based water monitoring solutions depend on terrestrial communication networks (e.g., GSM, LTE, Wi-Fi, or LoRaWAN) [7], which are unavailable in

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remote or rural regions. This connectivity gap limits large-scale deployment and continuous data acquisition from isolated monitoring sites.

Satellite-based IoT communication provides a promising alternative, offering global coverage, low power consumption, and resilience to infrastructure failure. When integrated with autonomous, multi-parameter sensors and edge intelligence, it enables continuous, real-time environmental observation without dependence on ground networks.

This research aims to design and implement a **real-time water quality monitoring system** that integrates IoT sensing with **satellite communication** for data transmission. The system targets remote and infrastructure-deficient sites such as rivers, reservoirs, aquifers, and abstraction points. Its main contributions are:

1. Development of a low-power, modular, multi-sensor node for continuous water quality measurement.
2. Integration of satellite-based IoT connectivity for global data transmission.
3. Implementation of local data processing and anomaly detection for event-driven alerts.

1. Related Works

Lakshmikantha et al. (2021) proposed an Internet of Things (IoT)-based smart water quality monitoring framework that continuously measures key water parameters such as pH, turbidity, conductivity, temperature, humidity, and carbon dioxide concentration in real time. The system utilizes Arduino microcontrollers integrated with various sensors to acquire data, which is then transmitted to a cloud server for further analysis and visualization. Their review of prior research highlights that IoT-enabled architectures significantly enhance real-time monitoring capabilities while reducing cost and manual intervention compared with conventional laboratory-based methods. The developed prototype demonstrated effective classification of drinkable and non-drinkable water samples, validating its practical applicability for sustainable environmental management and public health monitoring. Moreover, the authors suggested that future work should focus on integrating advanced sensors and wireless standards to improve scalability and responsiveness of such systems.[1]

Muthulakshmi et al. (2024) developed an IoT-enabled system for real-time monitoring of pond water quality parameters such as pH, turbidity, temperature, and dissolved oxygen. Using NodeMCU and ESP8266 microcontrollers, the system transmits sensor data to the cloud for continuous tracking and analysis. The study highlights how IoT-based monitoring enhances environmental management through automation, scalability, and timely detection of water quality changes.[2]

Alshammari et al. (2025) proposed an IoT-based smart water quality monitoring system that integrates cloud computing and machine learning for efficient real-time analysis. The system employs multiple sensors to measure parameters such as pH, turbidity, temperature, and dissolved oxygen, transmitting data to the cloud for processing and predictive modeling. Their approach demonstrates how combining IoT with data analytics enhances the accuracy, automation, and scalability of environmental monitoring systems.[3]

Shete et al. (2024) proposed an IoT-enabled real-time water quality monitoring method tailored for aquaculture applications. The system integrates pH, temperature, and dissolved oxygen sensors with an Arduino-based floating buoy unit for continuous, in-situ monitoring. Data are transmitted via GSM to a cloud platform, enabling correlation analysis among parameters to prevent fish mortality and ensure optimal pond conditions. The study demonstrated reliable performance with low maintenance costs and practical deployment for sustainable fish farming.[8]

In our previous work, Akpokodje et al. (2025) conducted a real-time environmental monitoring study of the Ystwyth River in Mid-Wales using AquaSonde multiparameter sensors. The sensors continuously measured key water quality indicators, including pH, dissolved oxygen, electrical conductivity, turbidity, temperature, and nutrient levels such as nitrate and ammonia. The AquaSonde AS-2000 probe was calibrated following ISO and APHA standards and deployed for two months in a protected housing to prevent sediment interference. Data were logged every 15 minutes and retrieved using the SondeLink software for analysis. Additionally, a web and mobile application built with Mapbox and Firestore was developed to visualize real time sensor data, providing stakeholders with accessible environmental insights to support catchment management and pollution mitigation.[9]

Kadrolli and Kalnoor (2024) reviewed the role of IoT and smart sensor technologies in remote sensing applications across healthcare and agriculture. Their work highlights how advanced sensors integrated with IoT networks enable real-time environmental and physiological monitoring, supporting applications such as precision farming, soil moisture assessment, disease surveillance, and remote patient monitoring. The authors emphasize that combining remote sensing data with IoT platforms enhances decision-making accuracy, operational efficiency, and large-scale data acquisition. They also discuss emerging trends, technological challenges, and opportunities for improving sensor robustness, data integration, and system interoperability within smart health and smart agriculture ecosystems.[13]

LoRaWAN-based architectures have shown promise for medium-range communication with low power consumption [4], yet their coverage remains geographically constrained. Similarly, Zigbee and Bluetooth technologies are limited to local network implementations. To overcome these limitations, researchers have investigated satellite-enabled IoT networks (Sat-IoT) for remote sensing applications, including precision agriculture and ocean monitoring.[9,13–15]

In the context of water monitoring, multi-parameter probes capable of measuring pH, dissolved oxygen, and turbidity have been integrated with IoT platforms for real-time reporting. However, these implementations still depend on ground-based connectivity. The integration of satellite communication with autonomous sensor nodes remains under-explored, particularly in freshwater monitoring.

This study addresses that gap by proposing a scalable, satellite-connected IoT water quality monitoring system with local intelligence, designed specifically for remote and infrastructure-deficient environments.

2. Materials and Methods

2.1. System Overview

The EMWAC system consists of two cooperating microcontroller units:

- **Arduino UNO R4:** Primary controller for sensor data acquisition, filtering, calibration, and LCD visualization.
- **Arduino MKR 1500:** Handles NB-IoT/LTE-M communication and satellite uplink via Lacuna Space.

Seven industrial-grade sensors are integrated:

1. pH (analog probe)
2. Dissolved oxygen (galvanic probe)
3. Electrical conductivity (EC probe with PT1000)
4. Temperature (PT1000)
5. Flow rate (Hall-effect turbine)

6. Ammonium ISE130
7. Nitrate ISE131

The system is powered by a 12.8 V, 6 Ah LiFePO₄ battery and a solar charging circuit.

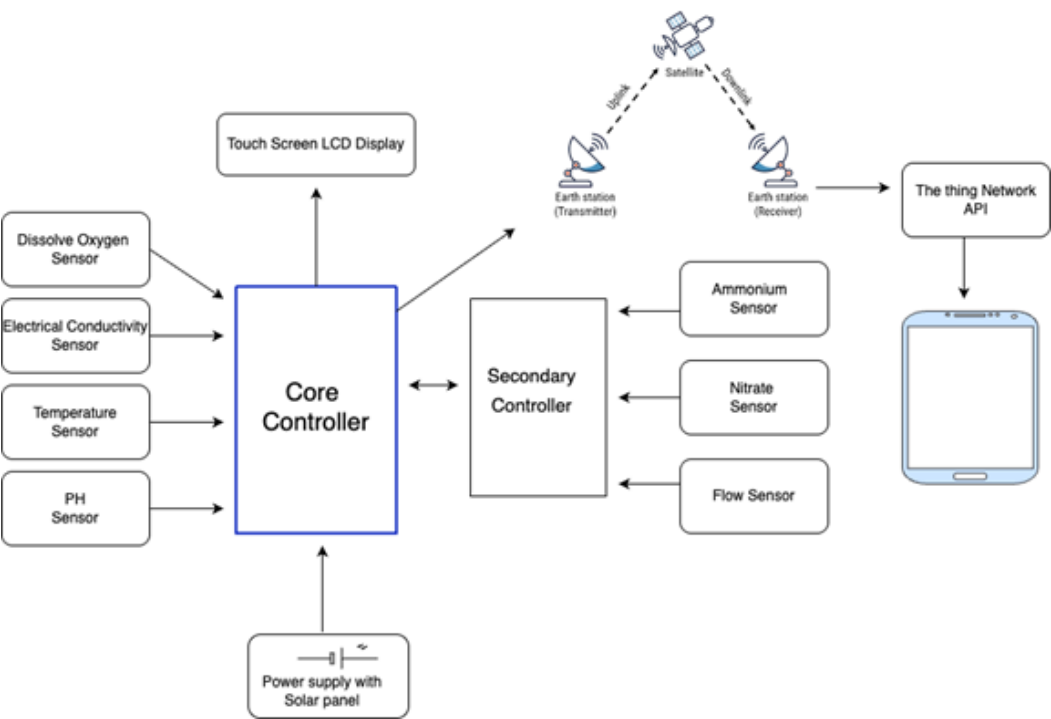


Figure 1. Block diagram of EMWAC sensor device

2.2. Hardware Architecture

2.2.1. Sensor-to-Arduino Pin Mapping

Sensor	Arduino Pin
pH	A0
EC	A1
Dissolved Oxygen	A2
Temperature	A3
Ammonium ISE	A4
Nitrate ISE	A5
Flow Rate	D2 (interrupt)

A TFT LCD module (ST7789V) provides real-time status and diagnostics.

2.3. Calibration Procedures

2.3.1. pH Calibration

A two-point calibration procedure is used with pH 4.0 and 7.0 buffer solutions. Commands such as `enterph`, `calph`, and `exitph` are issued through the calibration firmware.

2.3.2. Electrical Conductivity Calibration

Calibration uses a 1413 $\mu\text{S}/\text{cm}$ standard solution. The firmware updates the cell constant K after receiving a `C 1413` command.

2.3.3. Dissolved Oxygen Calibration

The DO probe requires:

- Filling the membrane cap with 0.5 mol/L NaOH

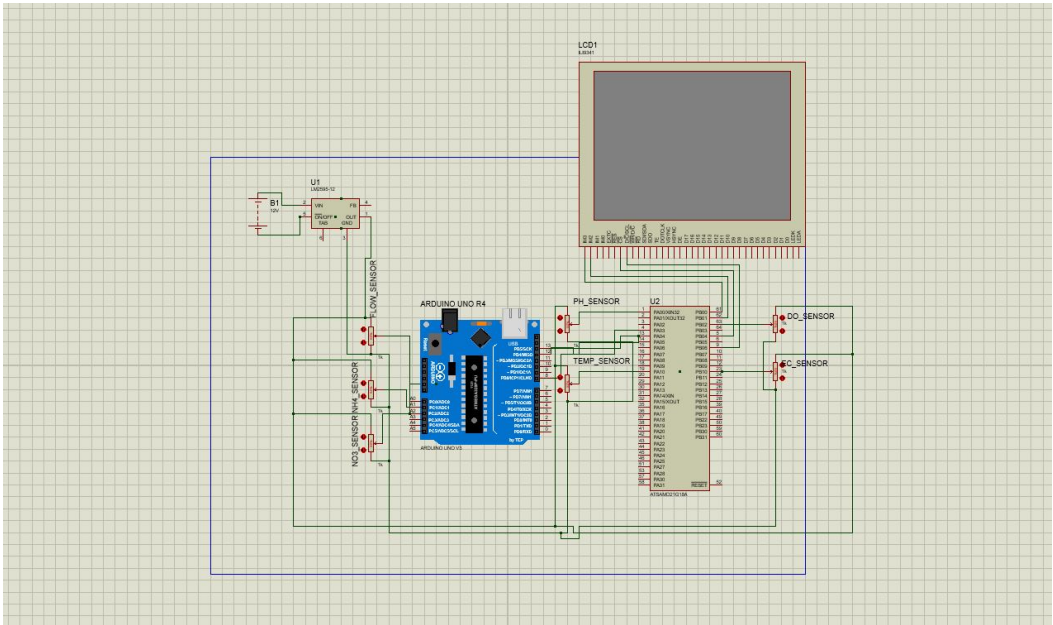


Figure 2. System schematic

- Single-point or two-point calibration in oxygen-saturated water
- Temperature compensation based on saturation curves

2.3.4. Ammonium and Nitrate ISE Calibration

Both ISEs are calibrated using two standard concentrations:

- 1 mg/L
- 100 mg/L

Ionic strength adjuster (ISA) is added for stability.

2.4. Communication Subsystem

The Arduino MKR 1500 handles NB-IoT/LTE-M transmission and satellite messaging. Lacuna Space’s LoRaWAN-over-satellite protocol is implemented following the official API. The satellite configuration includes:

- AES key setup
- Device address registration
- Country and geolocation settings
- Pass prediction for scheduling uplinks

2.5. Power Subsystem

The solar power system includes:

- 12.8 V LiFePO₄ battery
- Solar charge controller
- 5 V and 3.3 V DC–DC converters

3. Results

3.1. Multi parameter Sensors

All seven sensors produced stable, calibrated measurements. EC and DO accuracy improved significantly after temperature compensation.



Figure 3. Our multi- parameter sensor device

3.2. *Communication Reliability*

NB-IoT/LTE-M achieved over 95% packet delivery during testing. Satellite transmission was successful during Lacuna pass windows.

3.3. *Power Consumption*

Under solar charging, the system maintained a 60–90% state of charge during multi-day tests.

4. Discussion

The EMWAC system successfully demonstrates the feasibility of satellite-connected IoT monitoring in remote water bodies. The dual-MCU design provides modularity and resilience. Although satellite communication increases system cost and energy requirements, these are offset by reduced maintenance and global coverage capability.

5. Conclusion

This study has developed and validated a fully operational satellite-enabled IoT water quality monitoring system. The integration of industrial sensors, low-power microcontrollers, and satellite connectivity enables long-term deployment in remote environments. Future work will incorporate AI-driven forecasting, adaptive sampling, and integration with Earth observation data.

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Conflicts of Interest:

Abbreviations

The following abbreviations are used in this manuscript:

Appendix A

Appendix A.1

Appendix B

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